Development Specification

Sprint 2 – Intelligence Layer: Unanswered Question Detection

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1. Header

Project Name: Community Intelligence Dashboard
Sprint: Sprint 2 – Intelligence Layer
Story #2: Unanswered Questions List
Priority: High (Key Differentiator)
Owner: Community Intelligence Team
User Role: Community Manager

User Story:
As a community manager, I want to see a list of unanswered questions in my channels so that I can ensure no member feels ignored.

Sprint Goal:
Automatically detect questions in community channels and surface those that have not received responses.

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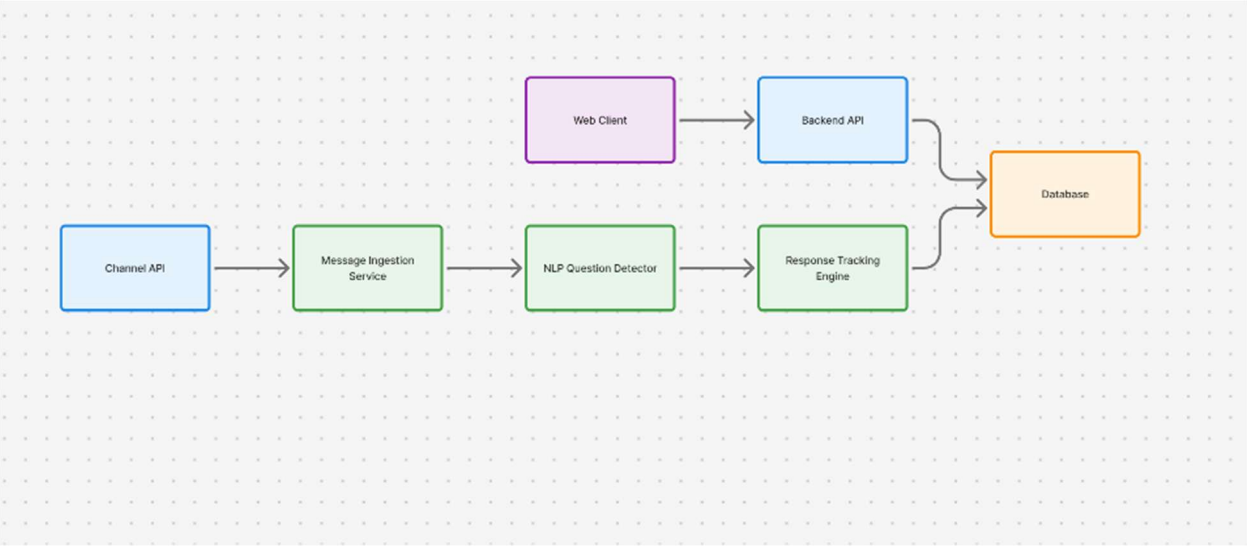
2. Architecture Diagram



Component Placement

Client (Web Dashboard)

- Community Manager UI
- Dashboard view
- Filters (Channel, Date, Status)
- Question list table

Server (Backend API Layer)

- Question Detection Service
- Response Tracking Engine
- REST API
- Authentication Middleware

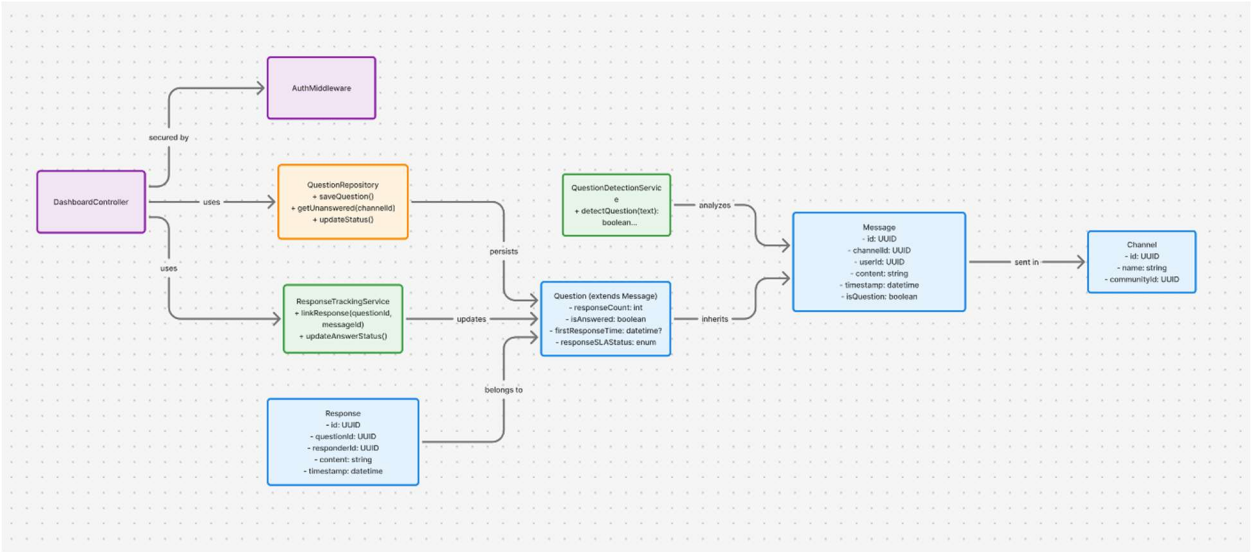
Cloud Infrastructure

- Database (PostgreSQL)
- NLP Service (internal microservice or cloud NLP model)
- Message Ingestion Service (listens to channel messages)
- Logging & Monitoring

Information Flow

1. Channel messages → Ingestion Service
2. Ingestion → NLP Question Detector
3. NLP → Marks message as Question (true/false)
4. Response Tracker → Monitors replies to question
5. Database → Stores:
    - Message
    - Question flag
    - Response status
6. Dashboard → Fetches unanswered questions via API

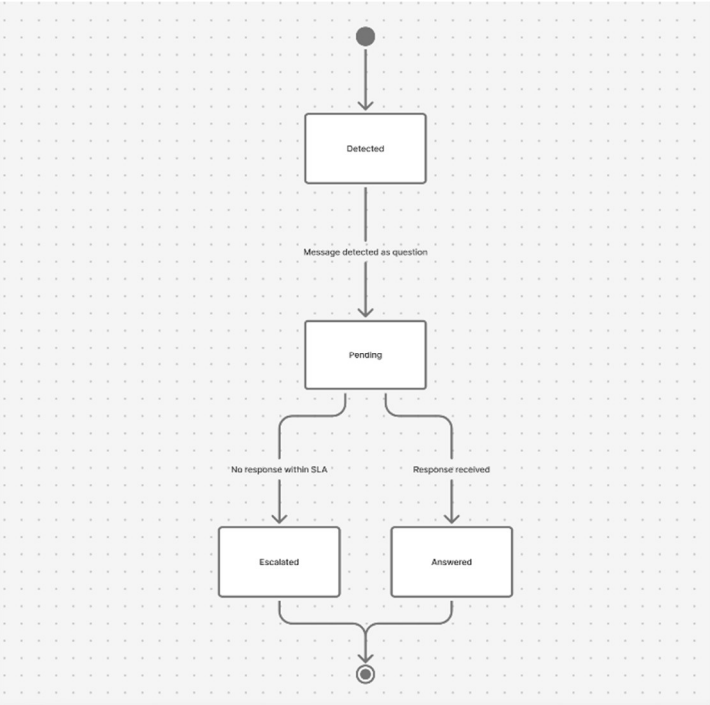
3. Class Diagram



4. List of Classes (Summary)

- Message
- Question
- Response
- Channel
- QuestionDetectionService
- ResponseTrackingService
- QuestionRepository
- DashboardController
- AuthMiddleware

5. State Diagrams



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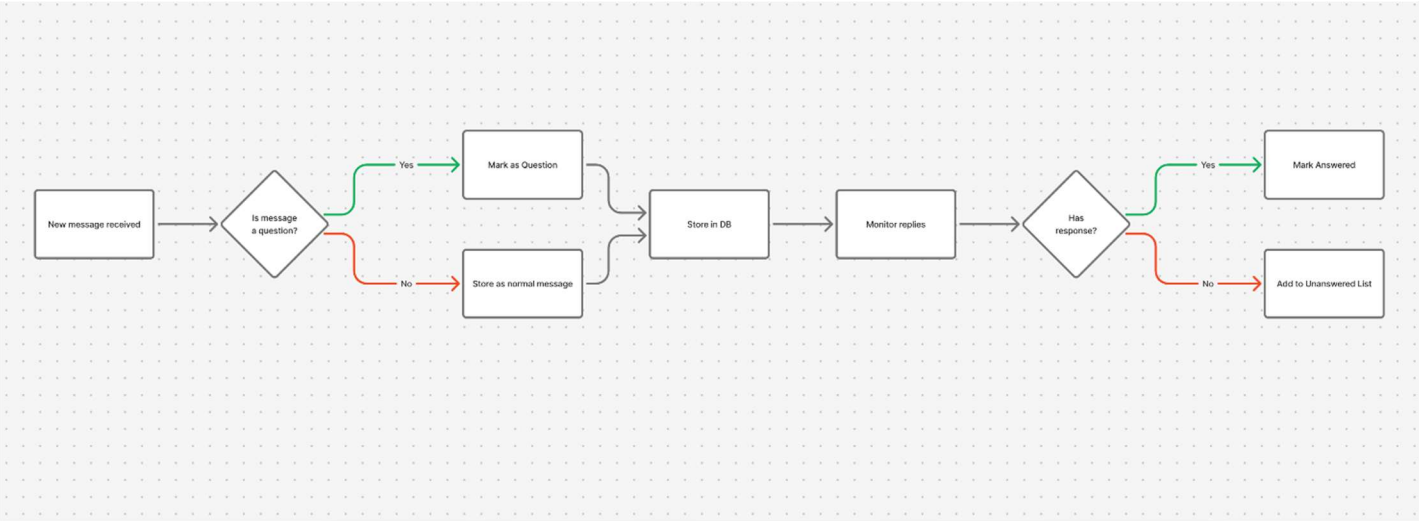
6. Flow Chart



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7. Development Risks and Failures

1. False Positives

- Statements ending in "?" but not real questions

- Mitigation: Confidence threshold tuning

2. False Negatives

- Questions without "?"
- Example: "Can someone help me"
- Mitigation: NLP model training

3. Threaded Message Complexity

- Reply may not directly reference question ID
- Mitigation: Use parent_message_id tracking

4. Performance Bottlenecks

- High message volume
- Mitigation: Async processing queue

5. Data Drift

- Model degrades over time
- Mitigation: Periodic retraining

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8. Technology Stack

Frontend

- React.js
- TypeScript
- Tailwind or Material UI

Backend

- Node.js (Express) OR Python (FastAPI)
- REST API

Database

- PostgreSQL

NLP Layer

- spaCy OR lightweight transformer model
- Hosted internally or via cloud service

Infrastructure

- AWS / GCP / Azure
- Docker containers
- Redis (optional for caching)

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9. APIs

GET /api/questions/unanswered

Query Params:

- channelId
- startDate
- endDate

Response:

```
[
  {
    "questionId": "uuid",
    "content": "How do I reset my password?",
    "channel": "general",
    "timestamp": "2026-02-14T12:00:00Z",
    "responseCount": 0
  }
]
```

POST /api/messages

Used by ingestion service to submit messages.

Internal Service API

detectQuestion(text: string)

Returns:

```
{
  "isQuestion": true,
  "confidence": 0.92
}
```

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10. Public Interfaces

Dashboard UI

Features:

- Table of unanswered questions
- Sort by:
  - Oldest first
  - Channel
  - Confidence score
- Filter by:
  - Channel
  - Date range
- Mark as resolved (manual override)

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11. Data Schemas

messages table

Column	Type
id	UUID
channel_id	UUID
user_id	UUID
content	TEXT
timestamp	TIMESTAMP
is_question	BOOLEAN

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questions table

Column	Type
id	UUID
message_id	UUID
is_answered	BOOLEAN
response_count	INT
first_response_time	TIMESTAMP

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responses table

Column	Type
id	UUID
question_id	UUID
responder_id	UUID
content	TEXT
timestamp	TIMESTAMP

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12. Security and Privacy

Authentication

- JWT-based auth
- Role-based access (Community Manager role)

Data Protection

- Encrypt in transit (HTTPS)
- Encrypt at rest (cloud DB encryption)

Privacy

- Store only message content required for detection
- No external NLP API unless compliant with data policy

Access Control

- Managers only see their community's channels

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13. Risks to Completion

Technical

- NLP model accuracy lower than expected
- Integration complexity with channel APIs

Time

- NLP tuning may exceed sprint capacity

Scope Creep

- Requests for sentiment analysis mid-sprint

Dependency Risks

- Message ingestion pipeline not stable
- Authentication system incomplete